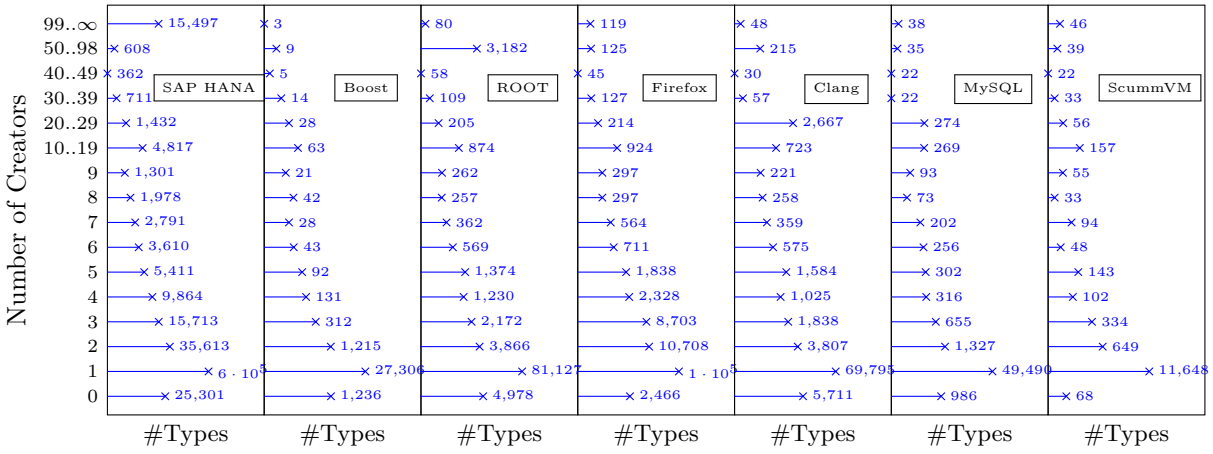


Figure 6.15: Histogram for number of object creators per type (x-axis in log-scale). E.g., for Boost, the search phase finds no creator in 1 236 cases and exactly 1 in 27 306 cases.

The most frequent cardinality is 1 due to the presence of DC in small classes. Table 6.5 shows the distribution of DC in different size groups of classes. It is rather common for small classes to have a DC. Large classes often do not have default constructors, but multiple creators. This indicates that our approach is more effective for large classes.

At least 6 % to 20 % of all object types and 16 % to 38 % of all large classes have more than 1 creator. This confirms the relevance of the OCP.

Answer RQ17

The search phase finds creators for 93 % to 99 % of all object types. The most frequent result is a single creator. For at least 6 % to 20 % of all object types, there exists more than one creator.

6.2.6.3 RQ18 Existence of Solutions

RQ18 What is the fraction of functions in studied C++ projects for which our approach can successfully find solutions for all arguments?

In practice, it is important to create all arguments of a function, which may require instantiating multiple different object types. Therefore, we switch our focus from objects to functions. In this section, we define a dependency graph (DG) of a function f as a union of dependency graphs for the types of each (required) parameter of f. Analogously, a valid sequence/-solution for f is the union of the respective concepts over all parameters of f. In other words, we introduce ana artificial root node, consider f as an creator for this root and apply our approach accordingly.

Functions: Our Clang plugin reports all functions generated by the compiler. This includes static functions, object member methods, lambdas, and each function template instantiation. Table 6.4 presents the results.

Size of Dependency Graphs (DG) and Solutions: We apply our approach to each function and obtain a dependency graph DG and a solution S. Fig. 6.16 shows for all functions in each project the sizes of DG and S. Table 6.6 presents the percentage of functions $\%Solved$ where our approach finds a solution and, over all functions, the average of the solution size $\overline{|S|}$, the graph size $\overline{|DG|}$, and the number of parameters $\overline{|args|}$.